TASK 1 FINAL REPORT

Modular Battery Management Engine Design

Platform: ESP32 + Wokwi

Architecture: Scalable array-based BMS analysis

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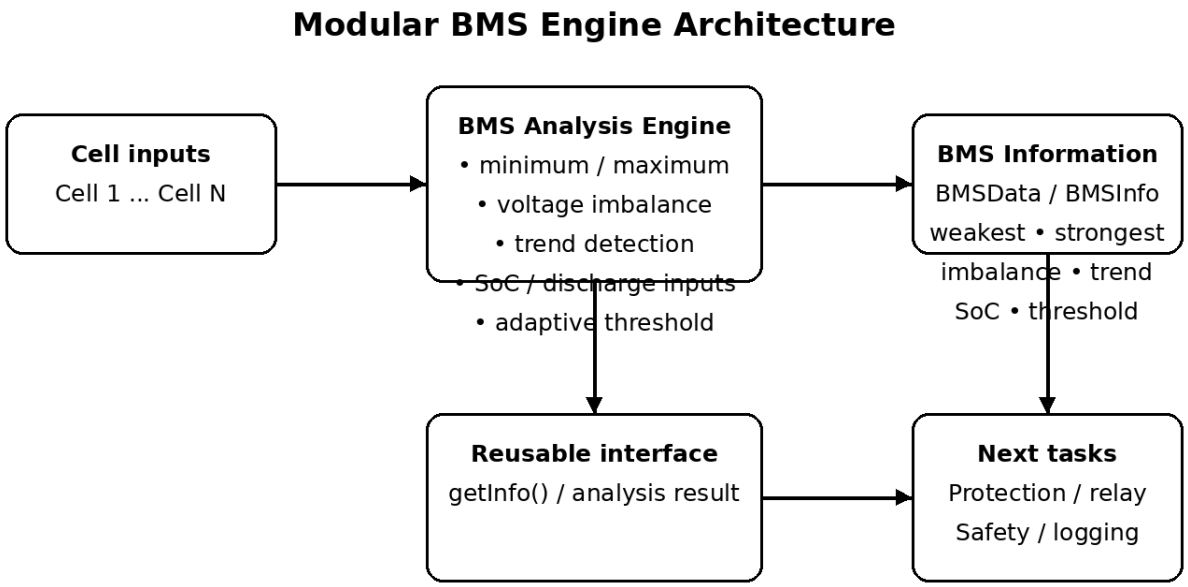


Figure 1. Modular BMS architecture and reusable analysis interface.

1. Introduction

This project implements a modular Battery Management System (BMS) analysis engine for an array of battery cells. A single compile-time constant, NUM_CELLS, controls the number of cells processed by the same algorithm. Four simulated cell voltages are provided by potentiometers on ESP32 ADC inputs, with additional potentiometers representing SoC and discharge rate.

The engine identifies the weakest and strongest cells, calculates voltage imbalance, tracks its trend, and calculates an adaptive imbalance threshold from SoC and discharge rate. A shared BMS data structure allows later protection and safety tasks to reuse the analysis.

2. Objectives

• Array-based cell processing

• Single compile-time cell-count constant

• Weakest/strongest cell detection

• Voltage imbalance calculation

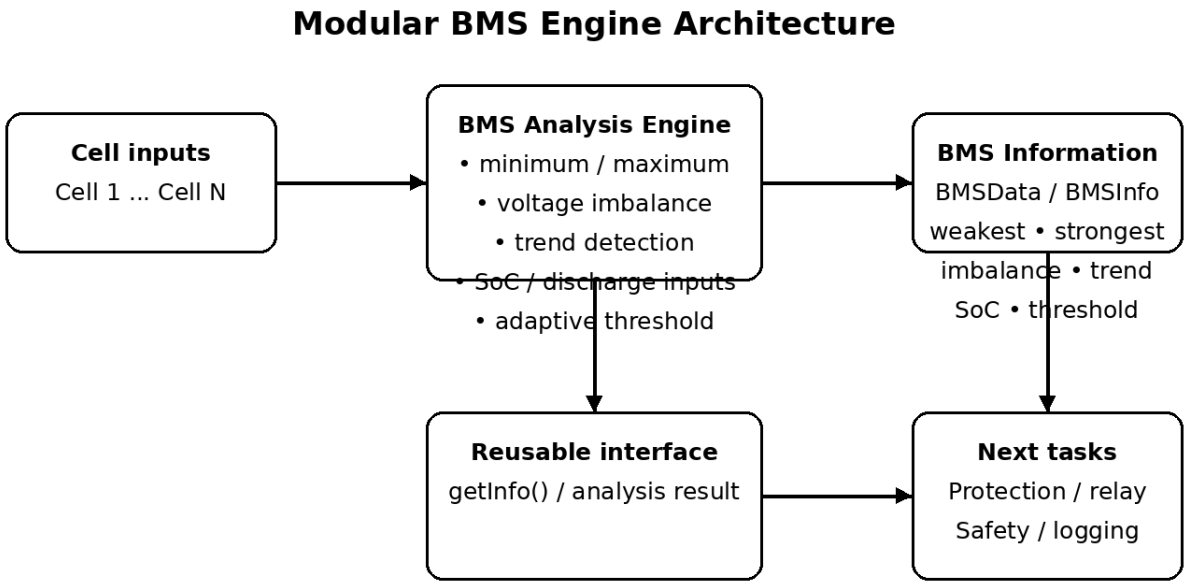
• Increasing/decreasing/stable trend detection

• Adaptive SoC/discharge threshold

• Reusable battery-information interface

• 4-to-16-cell scaling analysis

3. System Architecture



The analysis engine is separated from display and protection logic. Future tasks can consume the analyzed battery information without duplicating cell-analysis calculations.

4. Wokwi Hardware / Simulation Design

Input	Purpose	ESP32 pin
POT 1	Cell 1	GPIO 32
POT 2	Cell 2	GPIO 33
POT 3	Cell 3	GPIO 34
POT 4	Cell 4	GPIO 35
POT 5	State of Charge	GPIO 36

POT 6	Discharge rate	GPIO 39

Each potentiometer is powered from 3.3 V and ground, with its wiper connected to the corresponding ADC input. The simulation maps ADC readings to an illustrative 3.0–4.2 V cell-voltage range.

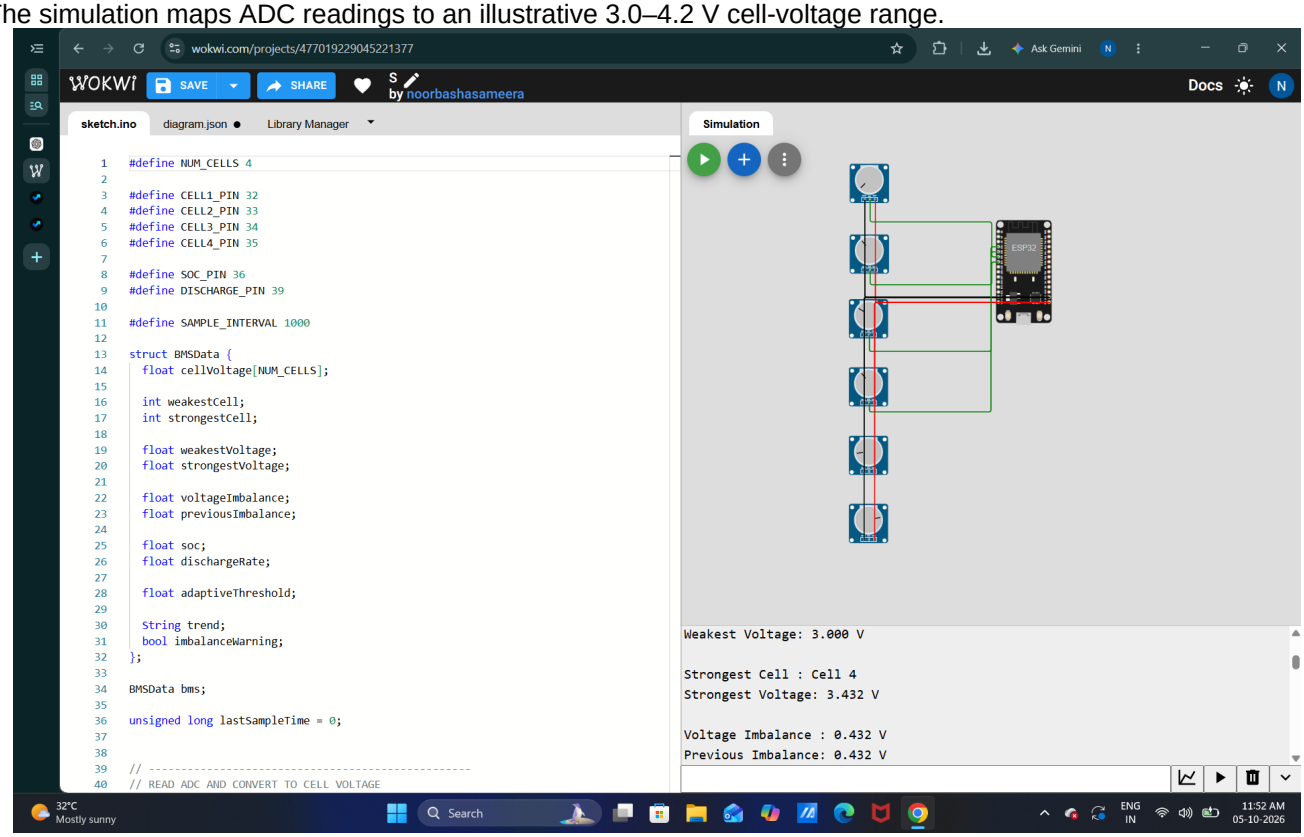


Figure. Weakest/strongest cell and voltage-imbalance test.

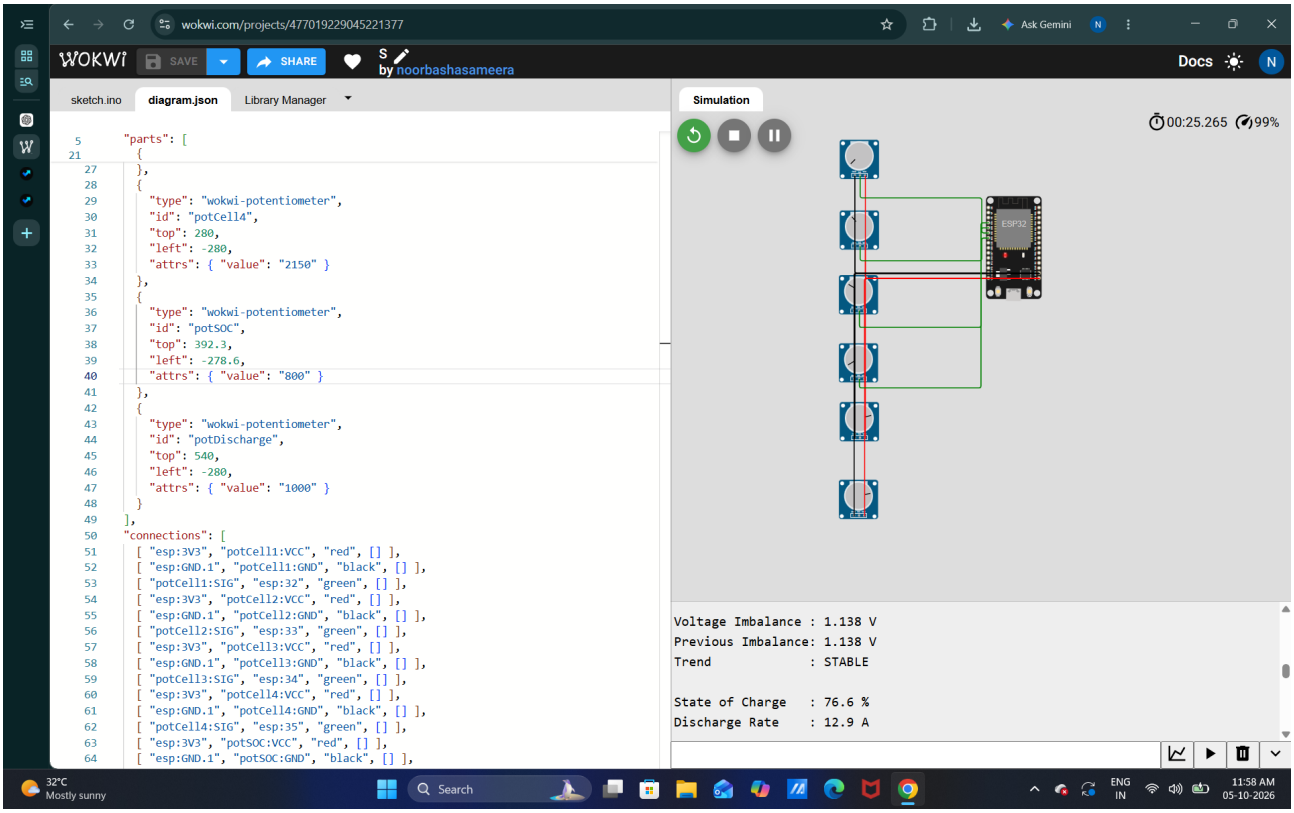


Figure. Wokwi circuit and six-potentiometer wiring in diagram.json.

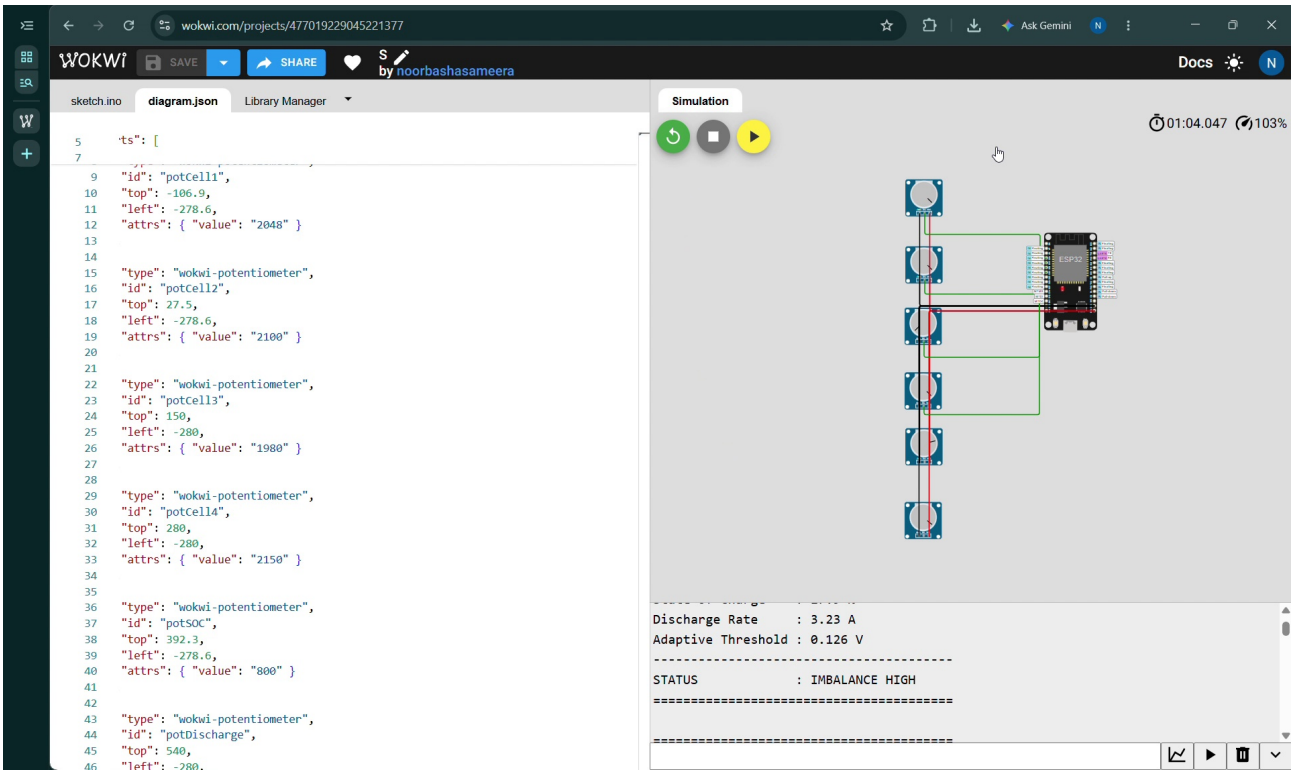


Figure. Adaptive-threshold and imbalance-high output evidence.

5. Software Design

The scalability mechanism is: #define NUM_CELLS 4 and float cellVoltage[NUM_CELLS]. All cell-processing loops use NUM_CELLS, so changing the constant to 8, 12, or 16 changes the array size and number of iterations without rewriting the analysis algorithm.

5.1 Data structure

BMSData stores cell voltages, weakest/strongest indices and voltages, current and previous imbalance, SoC, discharge rate, adaptive threshold, trend, and warning state. This creates one source of analyzed battery information.

5.2 Weakest/strongest algorithm

The engine scans the array once, retaining the minimum and maximum indices. Voltage imbalance = strongest-cell voltage – weakest-cell voltage.

5.3 Trend

Previous imbalance is stored before the current imbalance is calculated. A positive change above a small deadband is INCREASING, a negative change is DECREASING, and a small change is STABLE.

5.4 Adaptive threshold

The demonstration calculates a threshold from SoC bands and a discharge-rate term. These coefficients are simulation parameters and are not universal lithium-ion safety limits.

6. Test Results

Test	Action	Result / evidence
Balanced	Equalize four cell inputs	Small imbalance; stable/balanced expected
Weak cell	Lower Cell 3	Weakest Cell = Cell 3; demonstrated
Strong cell	Raise one cell	Strongest cell identified; demonstrated
Increasing trend	Gradually lower a cell	Capture final INCREASING screenshot
Decreasing trend	Raise weak cell	Capture final DECREASING screenshot

Adaptive threshold	Change SoC/discharge	Capture threshold-change screenshot

Observed evidence includes a case where Cell 3 is identified as weakest, Cell 1 as strongest, with 4.200 V – 3.000 V = 1.200 V imbalance. The same run reports an adaptive threshold and IMBALANCE HIGH.

7. Scalability: 4 to 16 Cells

Cells	Voltage-array memory	Comparisons	Complexity	Architecture change
4	16 bytes	3	O(N)	None
8	32 bytes	7	O(N)	None
12	48 bytes	11	O(N)	None
16	64 bytes	15	O(N)	None

With a 4-byte float, the cell-voltage array grows from 16 bytes at 4 cells to 64 bytes at 16 cells. The scan remains O(N), requiring 15 comparisons after initialization for 16 cells. Exact microsecond timing is not claimed because the implementation did not instrument cycle-level execution time.

8. Reusability and Interface

A clean final interface can expose the analyzed result through a method such as: BMSInfo info = bms.getInfo(); Future protection logic can then use weakestCell, strongestCell, voltageImbalance, adaptiveThreshold, and trend without duplicating the analysis.

9. Limitations and Engineering Notes

• Wokwi potentiometers are simulation inputs, not real cell measurement hardware.

• Real packs require suitable ADC scaling, protection, calibration and cell-specific safety limits.

• Adaptive-threshold coefficients must be validated for the actual battery chemistry and pack.

• Serial printing should be separated from the analysis engine in production firmware.

10. Conclusion

The Task 1 design demonstrates a scalable array-based BMS analysis engine. The same logic works for different cell counts through NUM_CELLS, identifies weakest and strongest cells, calculates imbalance, tracks its direction, and derives an adaptive threshold from SoC/discharge information. The design provides a reusable boundary for subsequent protection and safety tasks.

Appendix A — Submission Evidence Checklist

• Wokwi circuit

• Balanced output screenshot

• Weakest/strongest and imbalance screenshot

• Increasing trend screenshot

• Decreasing trend screenshot

• Adaptive-threshold screenshot

• Final sketch.ino and diagram.json

• 4-to-16-cell scaling analysis